

CellLineFinder: Integrated Multi-Omics Cell Line Selection

Advanced tools for
comprehensive cell line
analysis

How can we select the best cell line ?



- Cell line choice influences assay relevance, experimental robustness, and success in biopharmaceutical research.
- A data-driven problem improves planning and reproducibility.
- Integration of public datasets reduces manual data handling, accelerating timelines and freeing resources for analysis.



CellLineFinder: an integrated Data Platform for Targeted Cell Line Selection



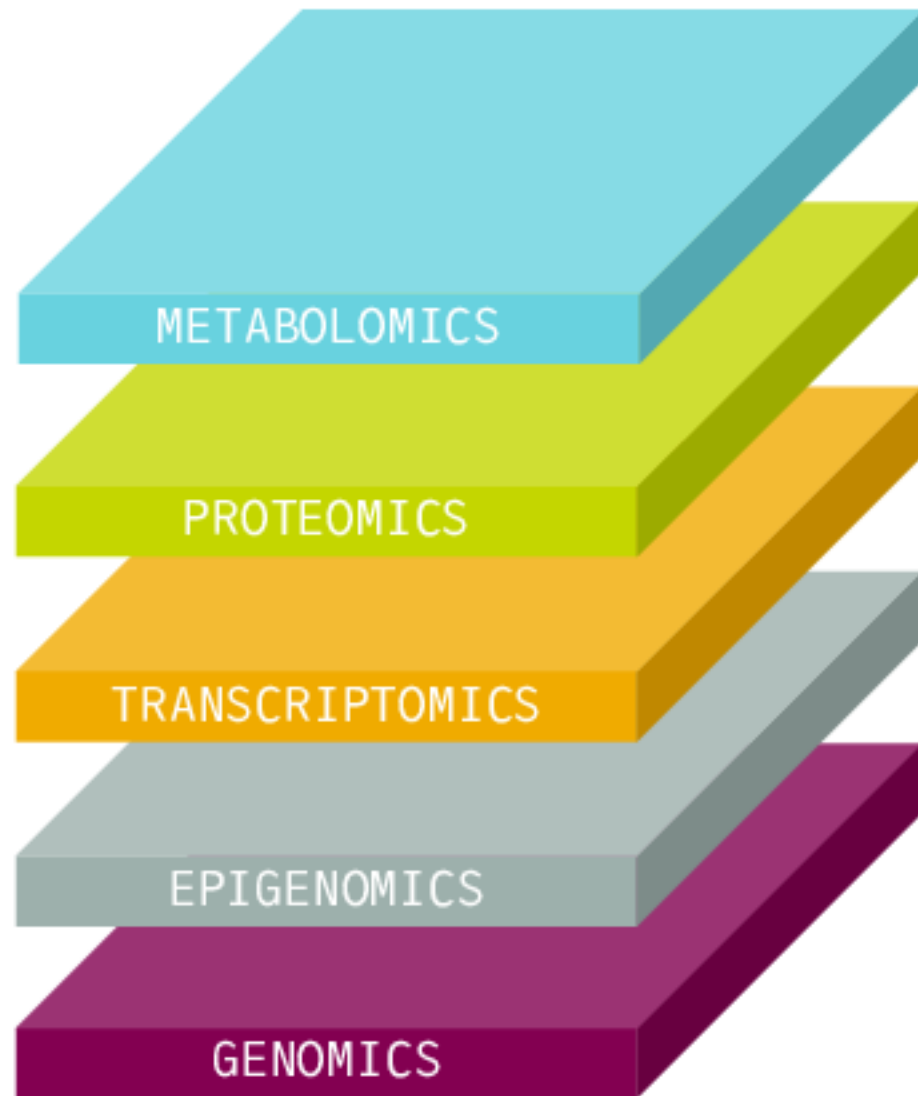
- Users specify molecular targets and exclusion criteria to identify optimal human cell lines for experiments.
- CellLineFinder unifies transcriptomic and proteomic data from major public resources into one harmonised queryable system.
- The platform provides ranked recommendations with confidence metrics and



Multi-omic characterization



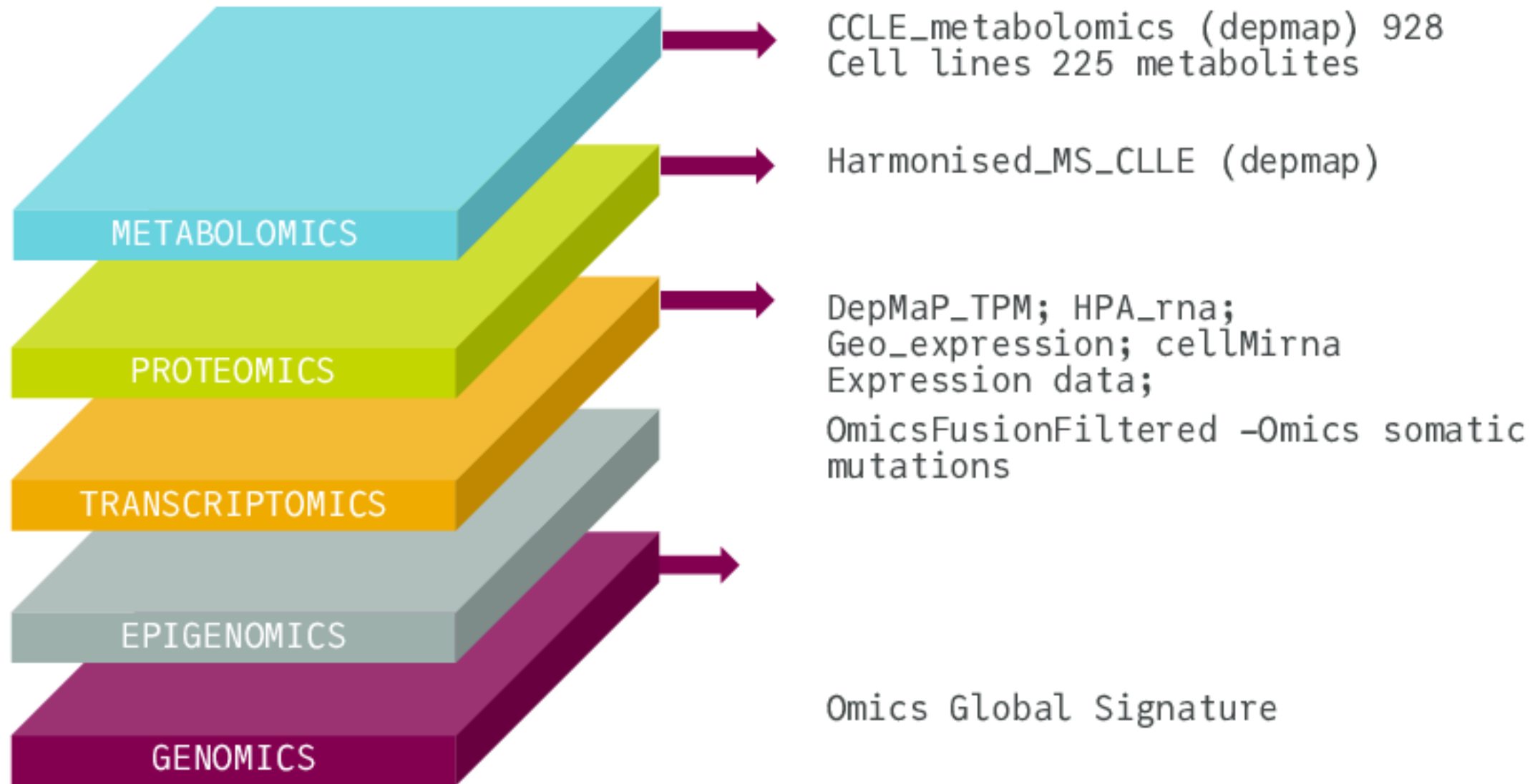
Which data are available ?



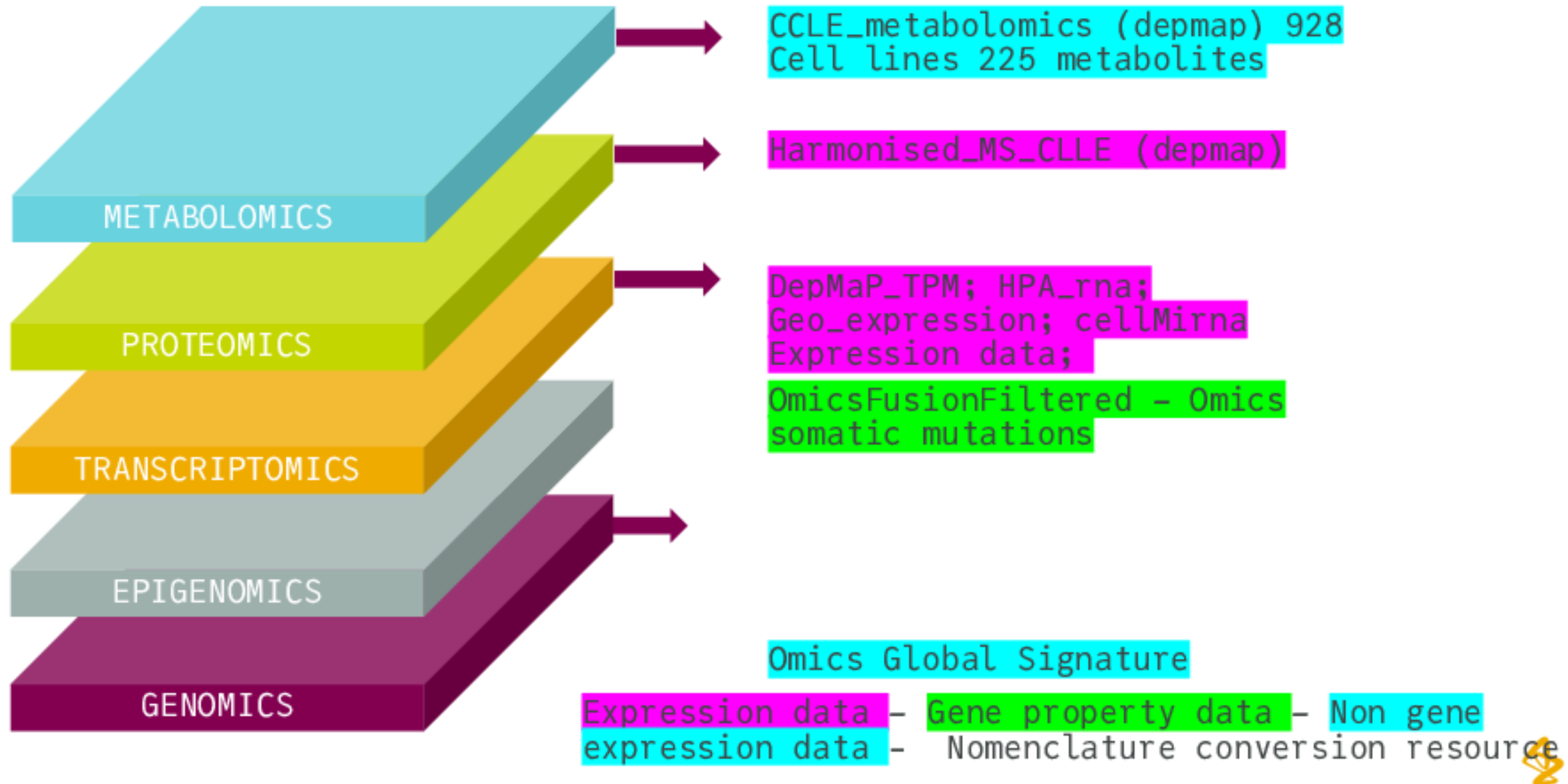
- Several publicly available datasets:
- <https://depmap.org/portal/>
- <https://www.proteinatlas.org/>
- <https://www.ncbi.nlm.nih.gov/geo/>
- <https://www.cellosaurus.org/>



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A gene fusion made by joining parts of two different genes. It may lead to the development of some types of cancer. For example, the *BCR::ABL1* fusion gene and protein are found in some types of leukemia.



Somatic variants develop in specific tissues due to DNA damage caused by aging or environmental factors like tobacco, UV radiation, and viruses.



RNA-seq data example

ensembl_gene_id	sym	gene_biotype	Cell-A	Cell-Z	Cell-B	Cell-C	Cell-D
ENSCGRG000001000001	ENSCGRG000001000001	Mt_tRNA	0	0	0	0	0
ENSCGRG000001000002	ENSCGRG000001000002	Mt_rRNA	627.054946	540.879894	584.166792	664.22124	630.150178
ENSCGRG000001000003	ENSCGRG000001000003	Mt_tRNA	0	0	1.169627	0	0
ENSCGRG000001000004	ENSCGRG000001000004	Mt_rRNA	2310.982648	2236.524475	1348.507175	1201.301754	1889.593781
ENSCGRG000001000005	ENSCGRG000001000005	Mt_tRNA	2.761283	9.195581	6.14054	6.476881	7.284374
ENSCGRG000001000006	ND1	protein_coding	4257.473243	4352.054048	3829.701538	3759.621038	5381.828847
ENSCGRG000001000007	ENSCGRG000001000007	Mt_tRNA	2.761283	1.946155	0	0.976673	0
ENSCGRG000001000008	ENSCGRG000001000008	Mt_tRNA	0	1.946155	1.169627	1.025506	3.450493
ENSCGRG000001000009	mt-Tm	Mt_tRNA	4.141925	2.919232	0	0.976673	0
ENSCGRG000001000010	ND2	protein_coding	5714.998844	5990.811758	5334.000654	4717.053519	6445.871769
ENSCGRG000001000011	ENSCGRG000001000011	Mt_tRNA	1.380642	0	0	0.976673	0
ENSCGRG000001000012	ENSCGRG000001000012	Mt_tRNA	4.141925	4.865387	8.187386	2.930018	4.600657
ENSCGRG000001000013	ENSCGRG000001000013	Mt_tRNA	136.68352	105.092353	92.400504	107.678154	86.262324
ENSCGRG000001000014	ENSCGRG000001000014	Mt_tRNA	13.806416	4.865387	4.678507	11.720071	4.856249
ENSCGRG000001000015	ENSCGRG000001000015	Mt_tRNA	0	0	0	0.976673	0
ENSCGRG000001000016	COX1	protein_coding	7775.033267	7369.701073	7908.376869	7550.013033	7633.859526
ENSCGRG000001000017	ENSCGRG000001000017	Mt_tRNA	56.606306	43.788481	40.936932	42.973594	52.204681
ENSCGRG000001000018	ENSCGRG000001000018	Mt_tRNA	0	0	1.169627	0.976673	2.428125
ENSCGRG000001000019	COX2	protein_coding	6270.58919	6442.184192	6150.996131	5827.19587	6315.28793
ENSCGRG000001000020	ENSCGRG000001000020	Mt_tRNA	0	0	2.339253	0	1.214062
ENSCGRG000001000021	ATP8	protein_coding	6366.143516	6469.638437	5486.738204	5193.975747	6155.07984
ENSCGRG000001000022	ATP6	protein_coding	11062.23576	10553.71728	9502.897823	8627.332304	10217.17282



End to end workflow

User Input and Data Query

Scientists specify target gene(s) and exclusion criteria (gene) to query a knowledge graph for relevant molecular evidence.

Evidence Aggregation and Scoring

The system aggregates transcriptomic and proteomic data, calculates confidence scores reflecting data strength and consistency.

Similarity Metrics and Alternatives

Alternative cell lines with similar molecular profiles are identified to support robustness and backup selection.

Results Presentation and Export

Ranked results with explanations are presented for user review, filtering, and export for reporting or documentation.

